

Worksheet

Gradient, midpoint and distance

For the line segment joining each pair of points in the table below, find the gradient, the midpoint and the length (in surd form).

	Points	Gradient	Coordinates of the midpoint	Distance between the two points
1	(0, 0) and (1, 3)			
2	(1, 2) and (3, 7)			
3	(2, 2) and (5, 10)			
4	(0, 5) and (2, 13)			
5	(3, -2) and (6, 6)			
6	(-4, 0) and (3, 5)			
7	(7, 1) and (9, -7)			
8	(2, 5) and (-4, 11)			
9	(6, 2) and (-5, -8)			
10	(5, 0) and (0, 6)			
11	(-3, 7) and (1, -1)			
12	(-1, -4) and (-4, -9)			

Answers

	Points	Gradient	Coordinates of the midpoint	Distance between the two points
1	(0, 0) and (1, 3)	3	$\left(\frac{1}{2}, \frac{3}{2}\right)$	$\sqrt{10}$
2	(1, 2) and (3, 7)	$\frac{5}{2}$	$\left(2, \frac{9}{2}\right)$	$\sqrt{29}$
3	(2, 2) and (5, 10)	$\frac{8}{3}$	$\left(\frac{7}{2}, 6\right)$	$\sqrt{73}$
4	(0, 5) and (2, 13)	4	(1, 9)	$\sqrt{68}$
5	(3, -2) and (6, 6)	$\frac{8}{3}$	$\left(\frac{9}{2}, 2\right)$	$\sqrt{73}$
6	(-4, 0) and (3, 5)	$\frac{5}{7}$	$\left(-\frac{1}{2}, \frac{5}{2}\right)$	$\sqrt{74}$
7	(7, 1) and (9, -7)	-4	(8, -3)	$2\sqrt{17}$
8	(2, 5) and (-4, 11)	-1	(-1, 8)	$6\sqrt{2}$
9	(6, 2) and (-5, -8)	$\frac{10}{11}$	$\left(\frac{1}{2}, -3\right)$	$\sqrt{221}$
10	(5, 0) and (0, 6)	$-\frac{6}{5}$	$\left(\frac{5}{2}, 3\right)$	$\sqrt{61}$
11	(-3, 7) and (1, -1)	-2	(-1, 3)	$4\sqrt{5}$
12	(-1, -4) and (-4, -9)	$\frac{5}{3}$	$\left(-\frac{5}{2}, -\frac{13}{2}\right)$	$\sqrt{34}$